

Complexity analysis of sEMG and IMU-based kinematics for muscular fatigue monitoring in dynamic sports settings

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Abstract: Muscular fatigue can significantly impair joint stability, alter proprioception, and increase injury risk in multidirectional sports such as football. While superficial electromyography (sEMG) is widely used for fatigue detection, signal non-stationarity during dynamic movements limits the reliability of traditional methods. This study assessed the potential of complexity-based sEMG features in capturing myoelectric manifestations of fatigue during a dynamic task incorporating sport-related movement patterns, while examining variability in lower-limb joint kinematics. Forty-five football players were monitored using a synchronised wearable system comprising sEMG and Inertial Measurement Units (IMUs) during a running activity with unpredictable directional changes. Spectral sEMG features (Median Frequency and Instantaneous Median Frequency) were compared with complexity-based indices (Sample Entropy, SE; Permutation Entropy) across six muscles. Results suggest that SE may offer enhanced robustness against motion-related non-stationarity, exhibiting a progressive decrease across muscles. Specifically, the most consistent fatigue-related adaptations were observed in the soleus and the vastus medialis ($p < 0.001$). Concurrently, alterations in joint motion variability patterns were observed in the hip, knee, and ankle during the running cycle, which may reflect fatigue-induced adaptations. These findings suggest that SE is a promising indicator for monitoring neuromuscular adjustments in dynamic conditions. Furthermore, exploring the co-occurrence between complexity-based sEMG trends and IMU-derived

kinematics offers a valuable approach for contextualising adaptations in neuromuscular control during a challenging dynamic task. Algorithmic estimates suggest the theoretical feasibility of near-real-time computation, though full system validation in real-world sports environments remains a prerequisite for formally recommending coaching applications or injury-prevention strategies.

Keywords: muscle fatigue, surface electromyography, sample entropy, injury prevention, dynamic motor control, joint kinematic variability

1. Introduction

Muscular fatigue is a neuromuscular condition that manifests in a reduction of the force or power production capabilities [1]. When fatigue occurs, the electrical and mechanical characteristics of the muscle change[2]. The electrical factors include changes in firing rates of action potentials and the number of recruited motor units [2]. The mechanical factors include the reduction in the force generated from both slow- and fast-twitch muscle fibres [1]. Fatigue develops gradually during sustained cyclic, repeating physical activity, starting to manifest while the task can still be sustained, until total inability to continue the task [3]. Beyond physiological alterations, muscle fatigue induces a reorganisation in motor strategies attempting to maintain accuracy and precision of voluntary movements [4]. These sensory-motor deficits result from integrated changes at both the central and peripheral levels, specifically involving the central nervous system, motoneurons and muscle afferents [5]. The consequence is an altered proprioception of body segments and degraded joint position sense [6-8]. In tasks involving multiple body segments, fatigue often necessitates a functional reorganisation of multijoint coordination to compensate for muscular performance decline [9]. Fatigue leads to a decrease in muscle synergy during movement and to consequent coordination alterations of the body segments, a reduction in degrees of freedom, or an overall loss of efficiency [9].

One of the primary roles of muscles is the stabilisation of joints, thereby reducing the mechanical load on passive structures such as ligaments and tendons [10]. Lower-limb muscles regulate both stiffness and dynamic joint stability during the preparatory and reactive phases of movement [10]. For example, effective dynamic stabilisation of the knee joint is essential for reducing joint loading during manoeuvres such as sidestepping [10]. During sports activities, the forces acting on the knee and ankle joints can exceed several times the body weight, making muscular action critical to avoid exceeding the physiological limits of the ligaments [10]. This is particularly important in sports involving frequent cutting and jumping, such as volleyball,

66 basketball, American football, and football (soccer) [10]. When muscles are fatigued, their ability
67 to stabilise joints can be impaired or delayed, placing excessive strain on ligaments and
68 increasing the risk of stretching or tearing [10, 11]. Muscular fatigue is a risk factor in both acute
69 and overuse injuries [11]. A study conducted by UEFA about injury incidence in professional
70 football players found that acute injuries such as strains, sprains and contusions are more
71 frequent toward the end of both the first and second halves of football matches, when athletes
72 experience higher cumulative fatigue [12]. A recent study [13], affirmed that muscular fatigue, if
73 accumulated over time, when high training volumes are associated with inadequate recovery,
74 acts as a critical driver of overuse injuries, exacerbating injury risk.

75 Considering football's global popularity and extensive participation at both amateur and
76 professional levels, the overall burden of football-related injuries represents a significant
77 concern. Ligament injuries and strains are the second most common type of injury in UEFA
78 professional football, both for men [12] and women [14]. Lateral ankle ligament injury and medial
79 collateral ligament injury of the knee are the first and second most frequent ligament injuries,
80 respectively [14]. Anterior Cruciate Ligament (ACL) injury is almost as frequent as medial
81 collateral ligament injury but has the highest burden for women [14]. It has been estimated that
82 females, independent of participation level, have a risk of sustaining an ACL 2.2 times greater
83 than males [15]. The increased risk has been attributed to a mix of anatomical, hormonal, and
84 neuromuscular factors, including increased knee valgus during dynamic loading tasks,
85 differences in hip and knee anatomy, and hormonal influences on ligamentous laxity. This carries
86 direct implications when aiming at designing an injury prevention program dedicated to females
87 [15]. Injury consequences are varied and include physical (e.g., long-term osteoarthritis,
88 permanent disability after a severe injury), psychosocial (e.g., fear of re-injury, emotional
89 distress) and financial impact [12, 14].

90 Surface electromyography (sEMG) is a non-invasive technique which uses electrodes
91 placed on the surface of the skin to detect muscular activity, enabling objective and real-time
92 monitoring [16]. sEMG has been broadly used for muscle fatigue analysis in several studies [16-
93 18], with the possible application of avoiding sports injuries [19]. It is well known that the changes
94 in muscle fibre propagation velocity and the firing rate of muscle fibres could affect the sEMG
95 power spectrum [16, 19, 20]. During sustained isometric muscle contractions, the power
96 spectrum shifts to lower frequencies. Consequently, the study of sEMG in its frequency domain
97 provides important information about fatigue status. In particular, the median frequency of the
98 Power Spectrum decreases under fatigued conditions and can serve as an index of muscular
99 fatigue. Simultaneously, the development of muscular fatigue during sustained isometric

contractions prompts recruitment of additional motor units to maintain the target force, increasing sEMG amplitude through the summation of active motor unit action potentials. Several time-domain features derived from sEMG amplitude, including but not limited to Root Mean Square (RMS), Mean Absolute Value (MAV), and Integrated EMG (iEMG), have been widely employed in the literature as indicators of muscular fatigue [21, 22]. However, the relationship between signal amplitude and the force generated within the muscle varies across different fatigue protocols, making time-domain features unreliable as standalone fatigue indicators [23].

When aiming for real-time monitoring of muscular fatigue during highly dynamic contractions, it is necessary to account for the increasing non-stationarity of the sEMG signal. Higher non-stationarity in the sEMG causes increased variability in time-domain and frequency-domain parameters for various reasons, collectively referred to as “dynamic factors” [24]. Such factors include recruitment and de-recruitment of active motor units in the proximity of the surface electrodes, changes in muscle shape, and movement of the innervation zone relative to the surface electrodes [1], but also the change in the number of active motor units, changes in force through the range of motion, and changes in fibre and muscle length [17, 18]. Under such conditions, time-domain features may change in competing directions, rendering their variations ambiguous with respect to the fatigue state [18]. Concurrently, frequency-domain features such as MDF are computed via Fast Fourier Transform, which assumes signal stationarity. This assumption is not met during dynamic contractions, where both the frequency content and the amplitude of the sEMG vary continuously over time, further limiting the reliability of these features in dynamic contexts [23].

Several techniques have been developed in search of a myoelectric parameter which is sensitive to the fatigue-induced changes in the sEMG but insensitive to the dynamic factors [24]. Time-frequency domain features are specifically designed to analyse signals whose frequency content changes over time, making them particularly suited to capturing the progressive spectral compression that characterises muscular fatigue during dynamic tasks [23]. Authors reported that sEMG features extracted from the time-frequency domain are preferable to previously mentioned time-domain and frequency-domain features when studying dynamic contractions [25, 26]. Cifrek et al. [27] used the instantaneous median frequency calculated from the spectrogram to assess muscle fatigue during cyclic dynamic contractions of the quadriceps muscle. Hayder et al. [20] transformed the sEMG signal into the time-frequency domain using the Short Time Fourier Transform to calculate the instantaneous mean frequency and instantaneous median frequency. Subsequently, the goal was to assess lower-limb muscle fatigue during 400-m running performed with three different pacing strategies. The limits of time-frequency

approaches include high variance, which requires time-averaging of instantaneous mean frequency estimates over consecutive cycles to sufficiently reduce variability. Specifically, methods based on time-frequency representations have demonstrated potential during cyclic contractions, where the kinematics remain consistent across consecutive cycles. However, when contractions are unconstrained and random, the high variance of the instantaneous frequency prevents reliable fatigue tracking. This precludes their applicability to unconstrained random dynamic contractions [24], the most common in sports.

Previous research emphasised that although time-frequency features provide valuable insights into muscle fatigue, the adoption of complexity-based indices may overcome the limitations of traditional spectral approaches [18]. Among the complexity-based features, Entropy quantifies the irregularity or unpredictability of a signal, with higher values reflecting greater complexity and reduced pattern repetition [16]. Different algorithms have been proposed in the literature to compute Entropy in sEMG signals. For example, Hernandez and Camic [28] investigated fatigue-induced changes in vastus lateralis sEMG during knee extensions, using Sample Entropy (SE), an algorithm that studies similarities in the signal, excluding self-matches. The authors reported reduced complexity, as indicated by lower SE, when fatigue occurred. Other authors analysed deltoid sEMG during dynamic lateral dumbbell lifts, demonstrating that Permutation Entropy (PE) could more effectively distinguish non-fatigued, transition-to-fatigue, and fatigued states compared to classical spectral features such as median frequency [29]. PE differs from SE by quantifying signal complexity through the ordinal patterns formed by neighbouring values rather than amplitude-based similarity, making it more robust to noise and well-suited for analysing highly dynamic and non-stationary biomedical signals [30].

Despite the extensive literature on sEMG-based fatigue assessment, no consensus has yet been reached on the most reliable indicators of neuromuscular fatigue under dynamic, non-stationary conditions that closely resemble real-world sport performance. This gap is particularly critical in high-intensity sports such as football, where repeated accelerations, decelerations, and unpredictable changes of direction impose substantial neuromuscular demands. The progressive onset of muscle fatigue may impair force production capacity, alter motor unit recruitment strategies, and disrupt sensorimotor integration, ultimately compromising joint stability and increasing injury risk. Identifying sensitive and robust markers of fatigue in controlled dynamic sport settings is therefore essential for advancing injury prevention strategies.

In this context, the present study aimed, first, to compare different sEMG-derived features in detecting lower-limb muscle fatigue during a dynamic fatiguing routine, characterised

by frequent and unpredictable changes of motion direction. It was hypothesised that complexity-based features would be more robust than features from the frequency and time-frequency domains in identifying fatigue.

A second objective was to investigate the effect of fatigue on the variability of lower-limb joint kinematic patterns, assuming that fatigue would modify joint motion variability, reflecting changes in the stability and adaptability of neuromuscular control mechanisms. Such alterations may indicate a reduced capacity to maintain coordinated movement patterns and to effectively respond to task demands during challenging dynamic activities.

2. Materials and Methods

2.1 Participants and Ethics

Forty-five healthy adult football players (37 males, 8 females; 22.6 ± 4.67 years old; 71.1 ± 9.8 kg; 175.9 ± 8.4 cm) were recruited to participate in a single acquisition session. Inclusion criteria were being between 18 and 35 years old [31]; complete recovery from past injuries; active participation in football: training at least twice per week on a regular basis. Exclusion criteria included females currently pregnant; any contraindications for training or competition participation; any systemic, rheumatic or neuromuscular condition that would limit task performance; current use of anti-anxiety medications, muscle relaxants and tranquillisers; proven allergies to physiotherapy adhesive tape. During the acquisition session, the participants were administered the AHA/ACSMH/Fitness facility Pre-participation Screening Questionnaire. If clearance was achieved, the participant received the consent form. Participants were not excluded because of their injury history, as this choice would result in a biased study population that is not representative of the athlete population, in which injuries are frequent [32]. At least 24 hours before their testing session, the participants had to refrain from any high level physical activity different from their weekly training (more than one hour of intense training such as heavy lifting), excessive coffee (more than two coffee cups or two espressos), and alcohol consumption (more than two liquor shots/long drink cocktails or two glasses of wine or two cans of beer). All participants were informed of the study's objectives, experimental procedure, and associated risks, and written consent was obtained from each participant before starting the acquisition session. The study was approved by the Luxembourgish National Research Ethics Committee (CNER) in accordance with the Declaration of Helsinki (202403/04 v2.0).

2.2 Experimental Design

2.2.1 Experimental Protocol

Participants were asked to wear sports clothes, shorts, and training shoes to perform the test during a single 1h 30-minute testing session. After signing the consent form, the participants were equipped with full-body setup Inertial Measurement Unit (IMU) sensors and sEMG electrodes on 6 dominant leg muscles. Dominant leg was assessed by asking the participant the preferred leg to kick the ball [33]. After wearing the sensors, the participants started a standardised warm-up presented by [31] including 2-minute jog, 5 bodyweight squats, 2 submaximal and 3 maximal countermovement jumps.

The warm-up was followed by a familiarisation session with the SKILLCOURT technology, with the *150m Random Run (RR)* task. The SKILLCOURT technology (movement concepts GmbH, Schweinfurt, Germany) is a device composed of a TV screen and a playing court, including 8 outer (square fields) target fields, where the players' position is continuously scanned by a 2D LIDAR (Light Detection and Ranging) combiner with a 3D depth camera system [34]. For this protocol, the court dimensions were 4 x 4 m. Despite its compact dimensions, the SKILLCOURT allows for testing high-intensity agility by accumulating running distances of up to 150 m per exercise through rapid changes of direction [35]. The device has been used to show its higher transferability to football-relevant performance when compared to change-of-direction training [35]. During familiarisation, participants executed the task at 60% of their maximal power. After checking the sensors' placement and readiness, the participants started the fatiguing routine.

The fatiguing routine was composed of 4 sessions of *150m Random Run*, separated by 30-second rest periods (refer to Section 2.2.2). Participants rated their Rate of Perceived Exertion before and immediately after each RR session using the Borg Rating of Perceived Exertion scale (RPE; 6 to 20), a subjective measure designed to assess an individual's perception of the intensity of the physical activity [36]. The sEMG and IMU signals were collected for the entire duration of the fatiguing routine. To mark the start of the resting periods, flags were manually added to the recordings, while the stop was automatically flagged after 30 seconds.

Muscular activity was recorded at 2000 Hz using a wireless sEMG system (Ultium EMG, NORAXON, Scottsdale, AZ, USA) from 6 dominant leg muscles: Rectus Femoris (RF), Vastus Lateralis (VL), Vastus Medialis (VM), Biceps Femoris (BF), Tibialis Anterior (TA) and Soleus (SOL). The participant's skin was carefully prepared to reduce impedance and improve the quality of the sEMG signal. Hairs were removed, if necessary, and dead cells, dirt, and sweat were eliminated by gently scrubbing the area with abrasive conductive gel (NuPrep, Weaver and Company, USA). After skin preparation, a pair of circular electrodes (Ag–Ag/Cl; Top Trace Ceracarta, Forlì, Italy, diameter 2 cm, inter-electrode distance 2 cm) was placed on the selected muscles as shown in **Figure 1a**. To maintain constant contact between the probe and the skin and to limit cable

movement that could introduce mechanical noise in the data, tape was used to fix both the probe and the cables. A layer of an elastic, cohesive band (Tiger Tapes, London, UK) was used to cover electrodes, probes, and cables to prevent them from detaching during testing due to high-intensity running and sweating (**Figure 1b**). An IMUs system comprising 9 sensors, including accelerometer, gyroscope and magnetometer, was utilised, sampling at 200 Hz (Ultium Motion, NORAXON, Scottsdale, AZ, USA). Sensors were placed on the participant's thorax (C7), lumbar spine (T12-L1), sacrum (L5-S1), thighs (frontal and distal half), shanks (along the tibia), and feet (metatarsus). The synchronisation of the two systems was done via software (myoRESEARCH MR3.20, NORAXON, Scottsdale, AZ, USA).

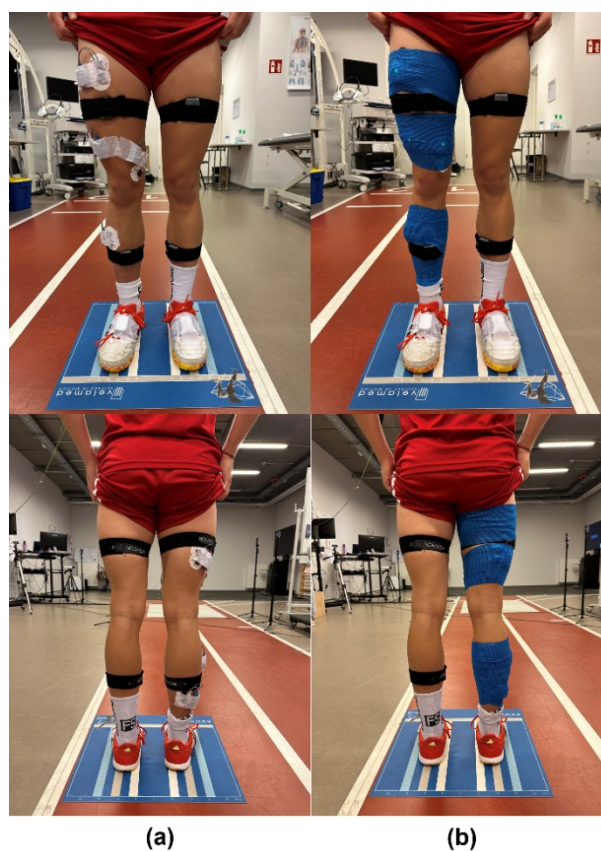


Figure 1: sEMG placement. (a) sEMG location, anterior and posterior views; (b) sEMG were covered by a layer of elastic cohesive band (in blue) to avoid detachment during testing due to high intensity running and sweating. Anterior and posterior views.

2.2.2 Fatiguing Routine (150m Random Run)

One of the available exercises in the SKILLCOURT technology is a routine consisting in four rounds of *150m Random Run* (RR), intermingled by 30-second rest. This reactive agility test couples directional changes with cognitive demands, using a task profile that elicits maximal

exertion [35]. In the RR task, participants need to hit target fields in a randomised sequence. On the screen, one of the outer target fields is highlighted in yellow, indicating the next target for the run. The system calculates the distance until 150 m is reached (**Figure 2**) [35]. The fatiguing routine is schematized in **Figure 3a**. The combination of multiple rounds of RR mimics football dynamics in terms of frequent changes of direction, periods of high-intensity activity interspersed by periods of low-intensity activity. Participants were instructed to finish the task as quickly as possible and were verbally motivated throughout the exercise. The fatiguing routine lasted an average of 450.59 ± 32.6 s (range: 404.35–518.07 s). Performance times generally increased across the rounds, with mean durations of 87.3 s, 90.36 s, 91.57 s, and 91.36 s for RR 1 through RR 4, respectively.



Figure 2: Illustration of the 150m Random Run (RR) exercise on the SKILLCOURT technology. The participant's goal is to hit the highlighted target fields in a random sequence, completing a total distance of 150m.

2.3 Data Analysis

Ten participants were excluded from the data analysis due to the following reasons: one presented IMU signal drift leading to out-of-range measurements, while the remaining nine experienced a foot IMU malfunction, which resulted in a failure to detect foot contact events. The final analytic sample comprised 35 participants (28 males, 7 females). Only the data acquired during the RRs, from the dominant leg, were analysed, while the data corresponding to the resting periods (highlighted in blue in **Figure 3b**) were ignored.

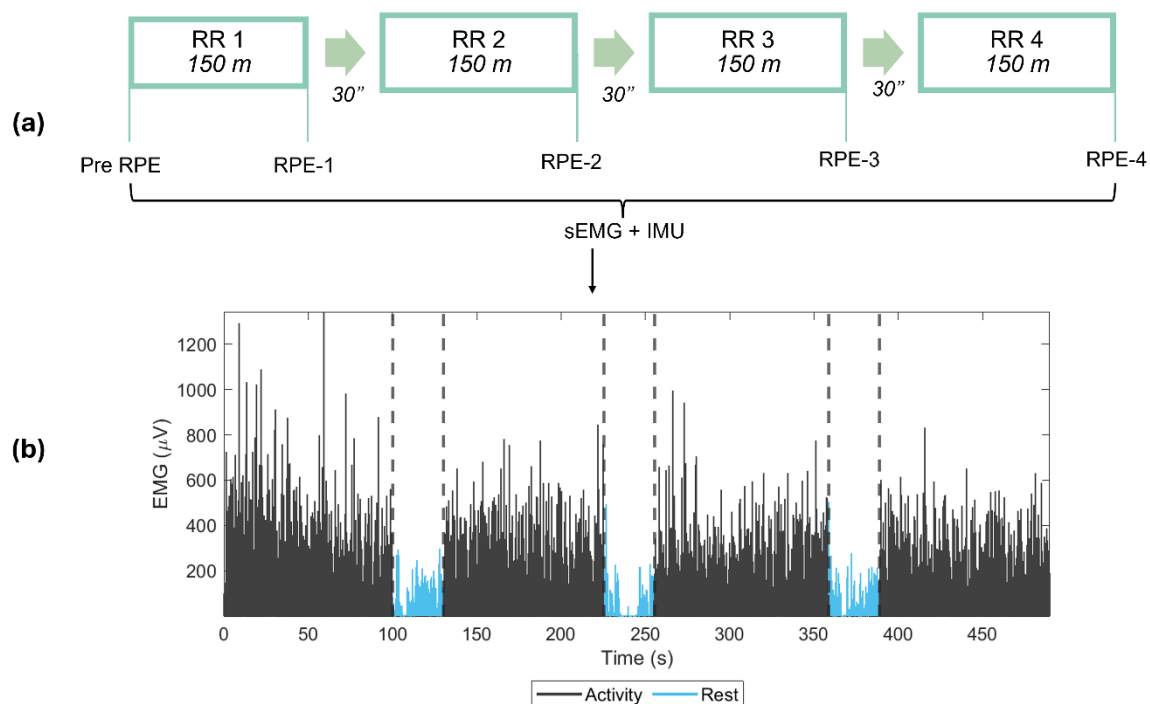


Figure 3: Diagram of fatiguing routine. (a) The fatiguing routine consisted of four rounds of 150m Random Run (RR) with a 30-second break between rounds. Participants rated their Rate of Perceived Exertion (RPE) before starting the routine and immediately after each RR. sEMG and IMU signals were collected throughout the fatiguing routine. **(b)** Example of sEMG activity collected during the fatiguing routine. Black signal corresponds to RR, and blue corresponds to resting periods. Data corresponding to the resting periods were ignored.

As a preliminary step, within the RR data, foot contact periods and flight periods of the dominant side were identified. This was achieved using myoRESEARCH software's automated detection algorithm based on the foot IMU sensor's angular velocity, angular acceleration, and linear acceleration signals. The software output is a binary time series in which the contact phase is encoded as one and the flight phase as zero. The initial point at which the binary series transitions from 0 to 1 was defined as Foot Strike (FS). Consequently, a Running Cycle (RC) was defined as the interval between two consecutive FS events of the dominant side. The foot IMU sensor position corresponded to the dorsum of the dominant foot, at the metatarsus level (see Section 2.2.1).

2.3.1 sEMG Analysis

The raw sEMG recordings were subjected to preprocessing, including the removal of noise and motion artefacts using a 2nd-order Butterworth bandpass filter (20-450 Hz), in accordance with recommendations for sEMG analysis during dynamic movements [20]. Following preprocessing, the signal was segmented for feature extraction in intervals of 300 ms, around the FS instants: 100 ms preceding the FS (pre-activation phase) and 200 ms following the FS (loading response phase). This window was selected because it encompasses the interval in which the most functionally relevant neuromuscular adjustments are known to occur during running [37, 38]. The pre-activation phase reflects preparatory muscle activity preceding ground contact [37], while the loading response captures the activation patterns that regulate impact attenuation and force production [37, 38]. The selection of these temporal windows aligns with methodological conventions reported in previous studies of running biomechanics [37-39].

An example of sEMG signals segmentation from the six analysed muscles is shown in **Figure 4**. The 300-ms windows considered for feature extraction are highlighted in green. The blue square-wave trace represents the binary foot-contact state derived from the foot IMU sensor, which enables identification of the FS (red dashed line).

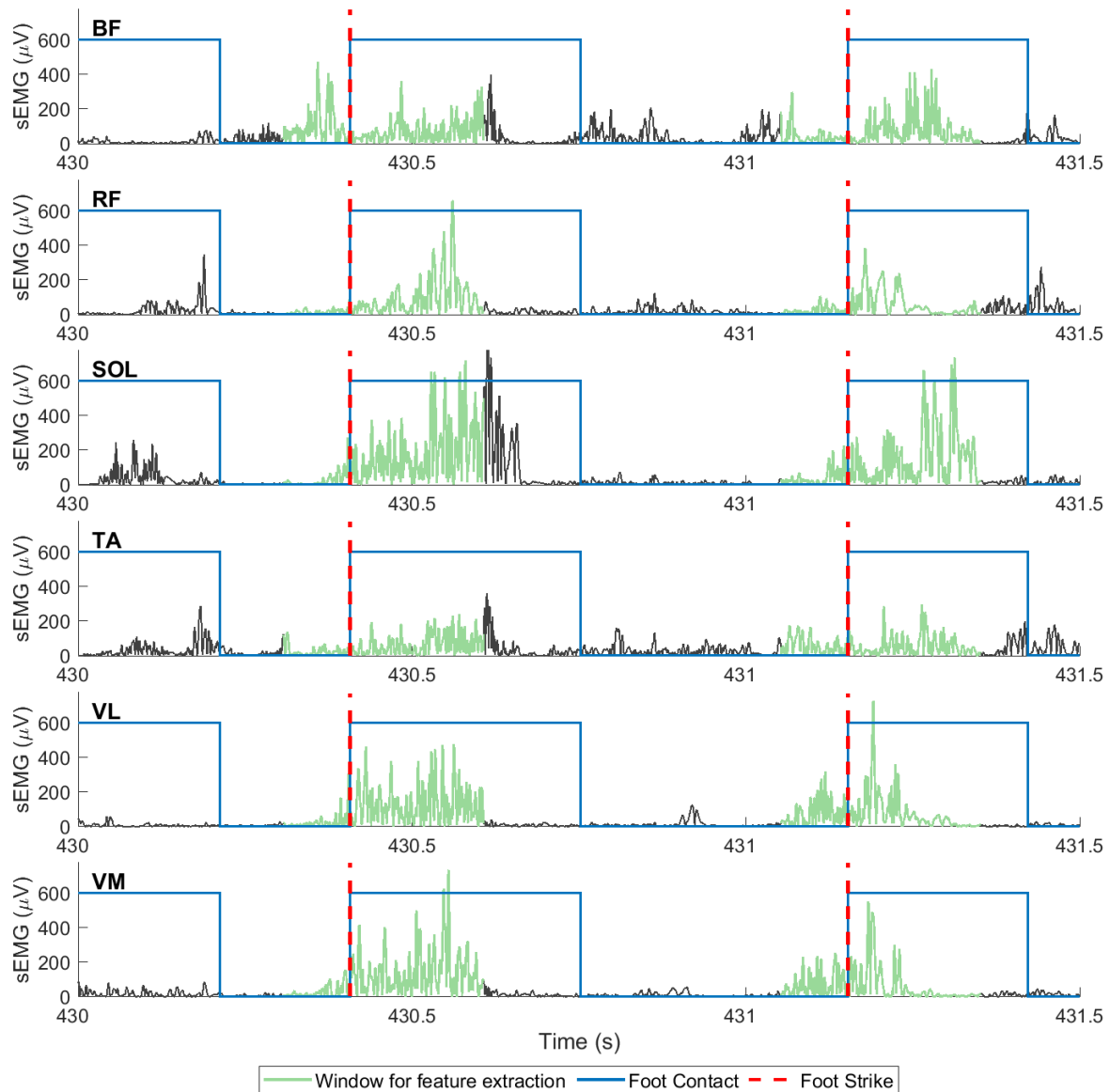


Figure 4: *sEMG signals of the six dominant limb muscles during the fatiguing routine.* The blue square-wave trace indicates the foot-contact binary time series, where the high state represents the contact phase, and foot strikes (FSs) are the initial point at which the contact signal rises above zero (red dashed line). The green-highlighted portions of the sEMG signals represent the specific time windows used for feature extraction. These 300-ms temporal intervals correspond to 100 ms preceding FS and 200 ms following FS. Muscles abbreviations: BF, Biceps Femoris; RF, Rectus Femoris; SOL, Soleus; TA, Tibialis Anterior; VL, Vastus Lateralis; VM, Vastus Medialis.

For each muscle, five features were extracted from each 300-ms window i. the Median Frequency of the power spectrum of the fast Fourier Transform (MDF: frequency at which the power spectrum is divided into two regions with equal area); ii. the mean value of the Instantaneous Median Frequency of the spectrogram obtained by the Short Time Fourier Transform algorithm

(mnIMDF) and iii. the maximum value of Instantaneous Median Frequency of the spectrogram (mxIMDF); iv. Sample Entropy (SE); v. Permutation Entropy (PE).

The selection of spectral features (MDF and IMDF-based metrics) and complexity-based features (SE and PE) was motivated by their established use in fatigue assessment during dynamic tasks [24-26, 28, 29] and by their ability to capture fatigue-related changes independently of amplitude modulation, which is particularly relevant in unpredictable multidirectional tasks.

The selected spectral and complexity-based features were defined as follows:

IMDF

The Instantaneous Median Frequency (IMDF) is an instantaneous feature obtained from transforming the sEMG signal from the time domain to the time-frequency domain. The STFT algorithm segments the sEMG signal into overlapping time windows of length L and applies a Fourier transform to each segment, resulting in a sequence of multiple power spectra. For each spectrum, the MDF is computed. The IMDF is the array of instantaneous MDF values computed across successive windows. The length L was set to 512 samples, and the window overlap to $L-1$ [27]. An overlap of $L-1$ was chosen to maximise temporal resolution of the IMDF, consistent with Cifrek et al. [27], enabling the tracking of spectral changes at each time point across the 300-ms window.

Sample Entropy

Sample Entropy is the negative natural logarithm of the conditional probability that two sequences that are similar for m (embedding dimension) consecutive points remain similar when one additional point ($m + 1$) is included, within a tolerance r [18]. Given a temporal series $X = [x_1, x_2, x_3, \dots, x_N]$, and a template vector $G(i) = [x_i, x_{i+1}, \dots, x_{i+m-1}]$, $n_i^m(r)$ is calculated as the number of vectors similar to $G(i)$, for all the vectors $G(j)$ excluding $i = j$ (self-matches). This leads to defining SE as [18]:

$$SampEn(m, r) = -\ln\left(\frac{A_i^{m+1}(r)}{B_i^m(r)}\right) \quad (1)$$

where:

$$A_i^m(r) = \frac{n_i^{m+1}(r)}{N-m-1} \quad (2)$$

$$B_i^m(r) = \frac{n_i^m(r)}{N-m-1} \quad (3)$$

In the present study, the tolerance r was defined as 0.2 times the standard deviation of each sEMG time series [28]. The embedding dimension m was set to 3, consistent with established practice in sEMG entropy analysis [40, 41]. Values of $m = 2$ or $m = 3$ have been reported as appropriate for sEMG signals, with Kang et al. [41] demonstrating consistent results across both values in leg muscle EMG. The use of $m = 3$ also ensured methodological consistency with the embedding dimension adopted for PE, enabling direct comparability between the two indices.

Permutation Entropy

Permutation Entropy is a complexity measure that quantifies the irregularity of a time series by analysing the relative frequencies of ordinal patterns within the signal. Instead of relying on amplitude values, PE captures the temporal ordering of successive observations, making it particularly robust to noise and amplitude scaling. For a time series $X = (x_i)_{i=1}^N$, the sequence is partitioned into overlapping vectors of length n , whose elements are ranked to form one of the $n!$ possible ordinal patterns[30]. Given p'_j , the relative frequency of each permutation pattern, the PE of order n is defined as:

$$PermEn(n) = - \sum_{j=1}^{n!} p'_j \log_2(p'_j) \quad (4)$$

In this study, PE was computed using an embedding dimension $n = 3$. This value falls within the recommended range of $m = [3 \text{ to } 5]$ for physiological signals [42]. This choice is consistent with recommendations that balance sensitivity to dynamic changes with the requirement for sufficient data length [30, 42]. The use of $n = 3$ also ensured methodological consistency with the embedding dimension adopted for SE.

2.3.2 IMU Analysis

To study body motion variability during the fatiguing routine, different angular features were extracted from the IMU signals. IMU signals were processed to compute nine joint angles across three anatomical planes: hip (flexion, external rotation, and abduction), knee (flexion, external rotation, and abduction), and ankle (dorsiflexion, inversion, and abduction). The angular data were analysed within the time interval of each RC. The Standard Deviation (SD) and Range (RA) were calculated separately for each angle (three for the ankle, three for the knee and three for the hip) during the contact and the flight phases of each RC. For each participant, the SD values

obtained during the contact phase were averaged separately for RR1 and RR4. Furthermore, the same averaging procedure was applied to the corresponding SD values calculated during the flight phase. An identical procedure was applied to the RA values.

2.4 Statistical Analysis

For each muscle and each of the five EMG features, the slope of the regression line computed from the values of successive 300-ms windows across all RR intervals was calculated individually for each participant. The resulting set of 35 slope values (one per participant) was then submitted to a one-sided t-test (H_1 : slope < 0) to assess whether the population average slope was significantly lower than zero. Specifically, the slope of the regression line fitted to the feature values was used as an index of fatigue progression. Negative slope values were interpreted as reflecting a progressive decline in the feature magnitude, and thus a potential manifestation of muscular fatigue, whereas a non-negative slope was interpreted as indicative of fatigue absence [20]. The normality assumption of residuals was examined, and for the cases where normality was violated, the non-parametric Wilcoxon signed-rank test was employed as an alternative. The significance level (α) was set at 0.05 and adjusted using the Bonferroni correction to control for multiple comparisons. P-values were considered significant when below $\alpha_{adjusted} = 0.0017$.

To assess joint kinematic variability, paired-sample t-tests were conducted separately for each of the nine joint angles, the two variability features (SD and RA), and the two phases of the running cycle (contact and flight phases), resulting in a total of 36 statistical comparisons. For each participant, the mean value obtained during RR1 was compared with the corresponding mean value obtained during RR4 to determine whether joint variability changed between the beginning and the end of the running task (H_1 : RR1 $\neq$ RR4). Normality assumptions were evaluated, and when violated, the non-parametric paired Wilcoxon signed-rank test was applied as an alternative. The significance level (α) was set at 0.05 and adjusted using the Bonferroni correction to control for multiple comparisons. P-values were considered significant when below $\alpha_{adjusted} = 0.0014$.

All statistical procedures were implemented in MATLAB R2023a (The MathWorks Inc., Natick, MA, USA).

3. Results

This section describes the data collected during the experimental protocol and presents the results of the statistical analysis.

3.1 Borg Rating of Perceived Exertion Data

Participants rated their perceived exertion before starting the fatiguing routine and immediately after each RR session using the Borg RPE 6 to 20 scale [36]. The average RPE increased progressively throughout the fatiguing protocol, rising from 8.4 ± 1.9 before exercise (Pre), to 14.1 ± 2.0 after RR 1, 15.6 ± 2.0 after RR 2, 16.7 ± 2.1 after RR 3, to 17.5 ± 1.9 after the last Random Run (RR4), indicating a gradual increase in perceived exertion as the protocol progressed.

3.2 sEMG Data

Five selected fatigue features were extracted from the preprocessed sEMG signal: MDF, mnIMDF, mxIMDF, SE, and PE. The slope of the regression line fitted to the feature values was used as an index of fatigue progression.

Figure 5 illustrates the mean slope (± 1 SD) of each sEMG feature across participants for each muscle. For each participant, the slope of the linear regression fitted to the feature values computed over successive 300-ms windows across all RR was used as an index of fatigue progression. The thick blue line represents the mean slope across participants, over the progression of the fatiguing routine, while the shaded area depicts ± 1 standard deviation. A negative slope indicates a progressive decline in feature magnitude over the fatiguing routine, interpreted as a manifestation of muscular fatigue, whereas a slope close to zero or positive suggests the absence of a fatigue-related trend. MDF, mnIMDF and mxIMDF showed a negative mean slope across RF and TA, while BF, SOL, VM and VL displayed mean slopes close to zero or slightly positive. SE showed negative mean slopes across all muscles. SOL and VM exhibited the most pronounced decreases, BF, TA and VL displayed small negative mean slopes, and RF remained close to zero. Regarding PE, all muscles showed positive mean slopes throughout the fatiguing routine progression, with BF, SOL, VM, and VL displaying the most pronounced upward trends, while RF and TA showed small positive mean slopes.

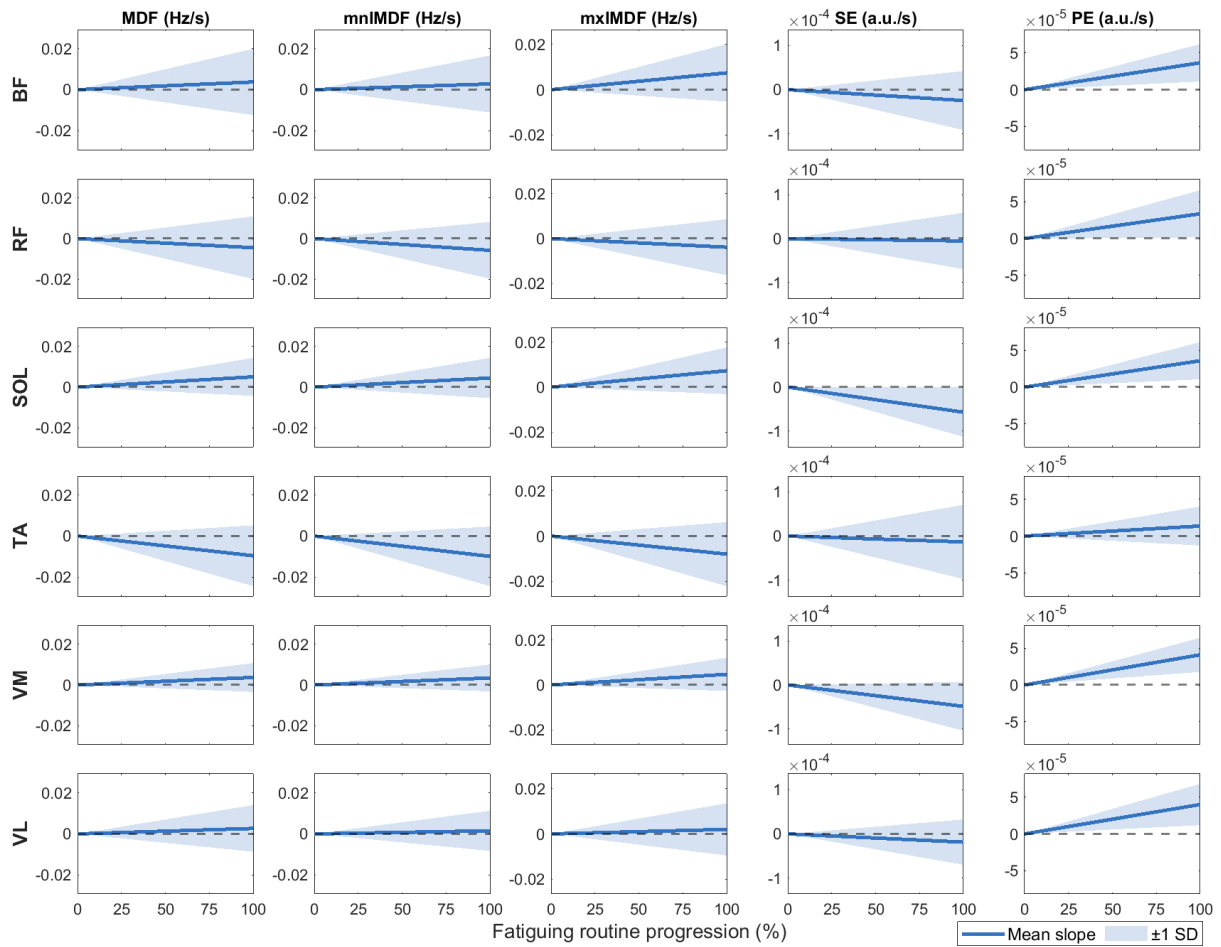


Figure 5: Mean slope and standard deviation of the fatigue features. From left to right: Median Frequency (MDF), Mean Instantaneous Median Frequency (mnIMDF), Maximum Instantaneous Median Frequency (mxIMDF), Sample Entropy (SE), and Permutation Entropy (PE). The figure shows the mean slope (tick blue line) and standard deviation (SD) (blue shaded area) computed across all participants. Muscles abbreviations: BF, Biceps Femoris; RF, Rectus Femoris; SOL, Soleus; TA, Tibialis Anterior; VL, Vastus Lateralis; VM, Vastus Medialis.

A summary of the statistical analysis of the five fatigue features is provided in **Table 1**. The analysis of MDF revealed a significant decrease in the TA slope ($p = 0.001$), while no significant changes were observed in the other muscles (all $p > 0.05$). Often, p-values were close to 1, indicating that the slopes were positive. Similarly, analysis of mnIMDF highlighted a significant decrease in the TA slope ($p = 0.001$), with no other muscles reaching significance ($p > \alpha_{\text{adjusted}}$) and some showing p close to 1. Regarding mxIMDF, TA showed a low p-value (0.003, Wilcoxon signed-rank), but after Bonferroni correction, none of the comparisons reached the adjusted threshold. Accordingly, no evidence of a systematic decrease in slope could be confirmed for this parameter. A similar trend is observed in this case, with some p-values close to 1.

For SE, statistically significant decreases were found in SO ($p = 4.14 \cdot 10^{-6}$), and VM ($p = 4.71 \cdot 10^{-5}$). BF and VL also showed low p-values, however not meeting the significance criteria after adjustment. Regarding PE, all muscles showed $p \approx 1$, suggesting that slopes were mostly positive.

Table 1. Summary of statistical analysis on sEMG data. The table contains the results for each feature-muscle combination, including the p-values, and the performed test. Statistics are computed on the distribution of participant-level regression slopes (N=35).

	Muscle	p-value	Executed test
MDF	BF	0.903	Wilcoxon signed-rank (one-tailed)
	RF	0.059	Wilcoxon signed-rank (one-tailed)
	SO	0.994	t-test (one-tailed)
	TA	0.001	t-test (one-tailed)
	VL	0.913	t-test (one-tailed)
	VM	0.995	Wilcoxon signed-rank (one-tailed)
mnIMDF	BF	0.913	Wilcoxon signed-rank (one-tailed)
	RF	0.005	Wilcoxon signed-rank (one-tailed)
	SO	0.974	t-test (one-tailed)
	TA	0.001	t-test (one-tailed)
	VL	0.823	t-test (one-tailed)
	VM	0.995	Wilcoxon signed-rank (one-tailed)
mxIMDF	BF	0.994	t-test (one-tailed)
	RF	0.087	Wilcoxon signed-rank (one-tailed)
	SO	0.999	t-test (one-tailed)
	TA	0.003	Wilcoxon signed-rank (one-tailed)
	VL	0.851	t-test (one-tailed)
	VM	1.000	t-test (one-tailed)
SE	BF	0.042	t-test (one-tailed)
	RF	0.322	t-test (one-tailed)
	SO	< 0.001	t-test (one-tailed)
	TA	0.166	t-test (one-tailed)
	VL	0.040	t-test (one-tailed)
	VM	< 0.001	t-test (one-tailed)
PE	BF	1.000	t-test (one-tailed)

	RF	0.999	t-test (one-tailed)
	SO	1.000	t-test (one-tailed)
	TA	0.999	Wilcoxon signed-rank (one-tailed)
	VL	1.000	t-test (one-tailed)
	VM	1.000	t-test (one-tailed)

3.3 IMU Data

Nine joint angles were extracted from IMU signals and the SD and RA were calculated separately for each angle during the contact and the flight phases of each RC. For each participant, the SD values obtained during the contact phase were averaged separately for RR1 and RR4. The same averaging procedure was applied to the corresponding SD values calculated during the flight phase. An identical procedure was applied to the RA values. For each participant, the mean value during RR1 was compared with the corresponding mean during RR4 to assess whether joint variability changed from the beginning to the end of the running task. Paired-sample t-tests were conducted separately for each of the nine joint angles, the two variability features, and the two phases of the running cycle.

The results of the statistical tests are presented in **Table 2**. The analysis highlighted a significant increase in SD and RA of knee external-internal rotation during the contact phase, as well as in ankle inversion-eversion throughout the entire running cycle.

At the hip level, Hip Abduction during contact showed an increasing trend, but not significantly.

At the knee level, Knee Flexion decreased during ground contact but was not considered significant. Knee External Rotation during the flight phase increased, but not significantly.

At the ankle level, dorsiflexion and abduction increased during contact and flight, respectively, but did not reach statistical significance. There were no significant differences in any other kinematic parameters tested.

Table 2. Summary of statistical analysis of motion variability using Standard Deviation (SD) and Range (RA). SD and RA were evaluated for 9 joint angles such as the hip flexion, external rotation, and abduction; knee flexion, external rotation, and abduction; and ankle dorsiflexion, inversion, and abduction. The table reports the p-value of the hypothesis tests comparing RR1 and RR4 values and trends for each of the nine joint angles, the two variability features, and the two phases of the running cycle.

			Standard Deviation		Range	
			p-value	trend	p-value	trend

Hip	Flexion	Contact phase	0.200		0.125	
		Flight phase	0.082		0.297	
	External Rotation	Contact phase	0.525		0.542	
		Flight phase	0.821		0.986	
	Abduction	Contact phase	0.005	↑	0.006	↑
		Flight phase	0.910		0.530	
Knee	Flexion	Contact phase	0.016	↓	0.027	↓
		Flight phase	0.278		0.102	
	External Rotation	Contact phase	< 0.001	↑	< 0.001	↑
		Flight phase	0.012	↑	0.013	↑
	Abduction	Contact phase	0.643		0.959	
		Flight phase	0.504		0.673	
Ankle	Dorsiflexion	Contact phase	0.254		0.586	
		Flight phase	0.017	↑	0.008	↑
	Inversion	Contact phase	< 0.001	↑	< 0.001	↑
		Flight phase	< 0.001	↑	< 0.001	↑
	Abduction	Contact phase	0.120		0.057	
		Flight phase	0.005	↑	0.002	↑

4. Discussion

The present study aimed to determine whether muscular fatigue could be detected during a highly dynamic fatiguing routine and if such fatigue would translate into altered lower-limb joint motion. The main findings were that complexity-based feature SE showed a clear decrease across the fatiguing routine and was therefore more sensitive to fatigue progression than traditional spectral features or an alternative complexity-based feature (e.g., PE). These other features displayed inconsistent behaviour under dynamic conditions. Moreover, fatigue was reflected in altered movement patterns: increased variability in knee external-internal rotation and ankle inversion-eversion, and reduced variability in knee flexion, indicating changes in joint control as fatigue accumulated. Together, these findings provide evidence that muscular fatigue during a routine characterised by repeated and unpredictable changes of direction affects both the electrical activity of the muscles and the biomechanics of lower-limb joints.

To the best of the authors' knowledge, no prior research has investigated the application of complexity features during multidirectional activities. On the contrary, existing fatigue studies are limited to cyclic, single-plane movements (e.g., treadmill running, cycle ergometer) [2, 20, 28, 29, 43]. In multidirectional and highly dynamic activities, the non-stationarity nature of the sEMG signal is even greater, and, as suggested by the present findings, many commonly used features are not suitable for detecting fatigue under such conditions.

The observed decline in SE is indicative of a loss of signal complexity. In line with previous literature, it can be hypothesised that this reduced complexity arises from altered peripheral mechanisms, such as decreases in action potential amplitude and velocity consequent to alterations in metabolic and enzymatic processes [18, 44]. Furthermore, fatigue-induced changes in motor unit discharge rates and recruitment strategies may shift neuromuscular activity toward more synchronised and stereotypical patterns, contributing to the overall reduction in signal complexity [18, 44]. The consistent decrease in SE across several muscles suggests that this complexity feature captures fatigue-related physiological changes even in highly dynamic tasks. In contrast, MDF, mnIMDF, mxIMDF, and PE frequently increased rather than decreased, despite participants reaching high levels of perceived exertion. When testing H_1 : slope < 0 , the p-values for these features were frequently close to 1, indicating that their slopes were systematically oriented in the positive direction, as also visible from the slope distributions reported in the figures. This discrepancy supports previous work indicating that spectral features are influenced by dynamic factors such as the contraction force [45, 46], repeated changes in muscle length, and successive recruitment and de-recruitment of motor units close to the electrodes [1]. These factors can hide the typical spectral shifts that occur in muscles during sustained physical activity. SE, on the other hand, consistently decreased across all muscles, making it the most suitable feature for analysing muscular fatigue in this context. Previous authors observed that SE is robust to changes in contraction intensity, whereas other complexity-based parameters exhibit a linear dependence on intensity [46]. This characteristic further supports the suitability of SE for fatigue assessment in uncontrolled environments and highly dynamic tasks.

The progressive decline in SE observed is further contextualised by the RPE trend. RPE increased progressively from 8.4 at baseline to 17.5 at RR4, a pattern temporally consistent with the SE decline. While a formal statistical relationship between RPE and SE was not tested in this study, the convergence of subjective perceived exertion with the SE signal supports the interpretation that the observed decline in SE reflects a progressive increase in muscular fatigue across the protocol.

550 Interestingly, the average trend observed in PE revealed a progressive increase across all
551 monitored muscles, in contrast to the loss of complexity usually associated with fatigue. Such
552 an increase might be cautiously interpreted as reflecting the asynchronous recruitment of new
553 motor units typical of prolonged dynamic tasks. A similar behaviour has recently been discussed
554 in the literature [47] suggesting that the direction of entropy trends might be critically dependent
555 on the specific embedding parameters selected or on the inherent non-stationarity of the motor
556 task. However, this topic remains largely unexplored, and further investigation is needed given
557 the current scarcity of literature on dynamic contractions.

558 Future work should examine whether the observed entropy differences are consistent
559 across different window lengths or gait phases, further validating the robustness of the approach
560 and elucidating their relationship to the physiological phenomena underlying the onset of
561 muscular fatigue.

562 IMU data analysis highlighted some differences in movement variability between the
563 initial and final stages of the fatiguing routine. The focus was placed on movement variability
564 because fatigue impairs the ability to execute movements in a consistent, stereotypical manner.
565 Changes in motor strategies were not evaluated, as the task involved unpredictable directional
566 changes, requiring uncontrolled strategies in response to the random stimulus. The correlation
567 between the observed kinematic changes and sEMG features was not explored, while qualitative
568 correspondence with EMG findings is presented here to contextualise the changes in movement
569 variability.

570 Several of the muscles showing fatigue trends play a key role in stabilising and controlling
571 the knee during directional changes. The VM, which exhibited a significant decline in SE, is
572 essential in knee flexion control [48]. When its function is impaired, individuals tend to limit knee
573 flexion to maintain stability, which is consistent with the reduced variability observed in this
574 plane. With reduced muscle function during heel strike, individuals seek stability by keeping the
575 knee extended, repeatedly performing the same knee flexion motion pattern with low variability
576 and a decreased flexion-extension range. The VL, which works synergistically with the VM to
577 stabilise the knee joint [49], also showed a decreasing SE trend, and reductions in the combined
578 stabilising capacity of the VM and VL may have contributed to the increased variability in knee
579 rotation. Furthermore, the BF, responsible for controlling knee external rotation [50],
580 demonstrated a decreasing SE trend, which aligns with the greater variability observed in knee
581 external-internal rotation. Overall, the highlighted alterations in lower-limb joint motion support
582 the notion that fatigue compromises the muscular action required to maintain joint stability
583 during multidirectional movements.

These observations suggest that at the final stage of the fatiguing routine, participants exhibited a reduced capacity to generate stable, efficient muscular output, reflecting a deterioration in neuromuscular control. The parallel analysis of complexity-based sEMG and IMU-derived kinematic metrics provides complementary perspectives on how fatigue manifests across both the muscular and kinematic domains. The present study examined complexity-based sEMG and IMU-derived kinematic metrics as complementary but independent measures of fatigue. Future work could explore their quantitative relationship through correlation or regression modelling, to better characterise how fatigue-driven changes in neuromuscular activation translate into altered joint motion variability during multidirectional tasks.

The fatiguing routine was intentionally designed to be highly dynamic, incorporating multidirectional and multiplanar movements. Its high intensity, with short resting periods and unpredictable, frequent changes of direction, required fast thinking and rapid motor planning. This design aimed to reproduce the neuromuscular and cognitive demands of intermittent sports, such as football, in which players must repeatedly respond to unpredictable stimuli.

While previous studies have attempted to replicate football-specific fatigue using laboratory-based protocols, they often relied on treadmill-based protocols that simulated running intensities in a single (forward) direction [51]. In contrast, the present routine introduced a more dynamic lab-controlled sport-settings model. It was designed to mimic sport-related movement patterns, with a specific focus on introducing frequent and unpredictable changes of direction, a crucial feature of multidirectional team sports. The protocol in a controlled laboratory space enabled precise data acquisition, high signal quality, and repeatability across participants. At the same time, it necessarily imposed a number of constraints that should be acknowledged when interpreting the findings. The testing area was limited to a 4 × 4 m surface, the total protocol duration was approximately 7.5 minutes, and the experiment was carried out in a controlled indoor environment, all of which diverge substantially from the spatial and temporal demands of a full match-play environment.

Future studies should expand this approach by integrating larger testing areas or longer protocols to examine how fatigue develops over extended durations comparable to full match-play. A larger spatial setup would more accurately replicate the locomotion patterns and distances covered during actual competition, while longer protocols would allow investigation of how fatigue accumulates and evolves over time, thereby better reflecting real match demands.

Additionally, the gender distribution of the present sample represents a notable limitation, as it precluded any sex-stratified analysis. With only 8 female participants, the sample

was underpowered to detect potential sex-specific differences in fatigue response. This is relevant given that documented differences in anthropometrics, muscle size, and fibre type distribution between sexes may influence exercise tolerance and potentially shape distinct neuromuscular responses to fatigue[52]. A more balanced gender distribution in future research would therefore allow for sex-stratified comparisons and a more thorough investigation of potential sex-specific adaptations. Identifying such responses could contribute to a better understanding of the higher prevalence of musculoskeletal injuries observed in women, and potentially inform more targeted injury prevention strategies [52].

5. Conclusion

The present study investigated the suitability of complexity-based sEMG features in detecting muscular fatigue during a multidirectional, high-intensity protocol. The results were consistent with the initial hypothesis, suggesting that SE may be a more robust and sensitive indicator of myoelectric manifestations of muscle fatigue than traditional spectral features and PE. The progressive loss of signal complexity, captured by SE, was associated with the onset of fatigue in SO and VM.

Furthermore, the analysis of joint angle variability confirmed the second hypothesis, revealing that muscular fatigue directly translates into altered lower-limb joint kinematics. The observed increase in Knee External Rotation and Ankle Inversion variability suggests a functional reorganisation of movement. The fatigue observed in muscles like VM impairs proprioception and joint position sense, leading to imprecise motor output. Consequently, the muscular ability to dynamically stabilise the lower limb during rapid changes of direction is compromised, significantly increasing injury risk.

This study demonstrated SE utility in dynamic, laboratory-controlled sports settings, designed to mimic sport-related movement patterns, with particular emphasis on unpredictable changes of direction, a crucial feature of multidirectional team sports. This expands previous investigations of EMG-based muscle fatigue assessment that have often been limited to studying fatigue in a very controlled motion, treadmill-based protocols, or running in a single direction[20, 24, 27-29, 51]. The focus of this research was to reproduce key kinematic features of multidirectional sport in a controlled laboratory environment, rather than to replicate the full spatial and temporal demands of real match play.

From a computational perspective, the ability to estimate SE over a short data window (300 ms) establishes a solid foundation for future real-time applications. SE computation on a 300-ms window (600 samples at 2000 Hz) required approximately 12.9 ± 0.5 ms (mean $\pm$ SD over

1000 runs) on a standard laptop (Intel Core i7-12700H, 2.3 GHz, 16 GB RAM), suggesting that near-real-time computation is theoretically feasible. However, full system validation including hardware integration, data transmission latency, and on-field testing remains outside the scope of this study and constitutes an important direction for future research.

Consequently, the potential impact of these findings lies in the development of real-time monitoring systems that alert coaches and athletes to suboptimal movement patterns as they occur. Future research should focus on validating this combined sEMG-IMU framework in actual competition environments and during full match play durations before coaching applications or injury-prevention strategies can be recommended. Such advancements are essential for optimising athletic performance while reducing the physical, psychosocial, and financial burden of sports-related injuries.

Data availability

Data are available via Zenodo at <https://doi.org/10.5281/zenodo.20847204>.

Declaration of generative AI use

During the preparation of this work, the authors used Grammarly generative AI assistance (Grammarly, Inc.) for assistance with grammar and proofreading. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the published article.

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